MNOP AS AN E_2-CENTRE AUTOMORPHISM, CHAIN-LEVEL S^3-FRAMING ON THE QUINTIC, AND THE CHIRAL-VOLUME ASYMPTOTIC

1. The MNOP substitution as an E_2-centre automorphism

Let X be a smooth compact Calabi–Yau threefold. Write $\mathrm{Perf}(X)$ for its perfect derived category, $\Phi_3^{\mathrm{FA}} : \mathrm{CY}_3\text{-cat} \to \mathrm{Fact}_{\mathrm{E}_3}$ the factorisation-algebra image in the sense of Costello–Gwilliam / Kontsevich–Soibelman with Kinjo–Park–Safronov orientation data, and set $\mathcal{F}_X := \Phi_3^{\mathrm{FA}}(\mathrm{Perf}(X))$. By Lurie (*Higher Algebra* 7.3.4.2), the E_2-centre $Z_{\mathrm{E}_2}(\mathcal{F}_X)$ is well-defined and carries a canonical E_2-algebra structure.

Three dualisable $\mathcal{F}_X$-modules are canonically attached to X:

- M_{DT}, the Donaldson–Thomas module (Thomas 2000), with partition function $Z_{\mathrm{DT}}(q) = \sum_n \mathrm{DT}_n(X) q^n$;
- M_{PT}, the Pandharipande–Thomas stable-pair module (PT 2009), with $Z_{\mathrm{PT}}(q) = \sum_n \mathrm{PT}_n(X) q^n$;
- M_{GW}, the Gromov–Witten module (Behrend–Fantechi 1997), with $Z_{\mathrm{GW}}(u) = \sum_g \mathrm{GW}_g(X) u^{2g-2}$.

Theorem 1.1 (Chain-level MNOP at the quintic)**.** *Let $Q \subset \mathbb{P}^4$ be a smooth quintic threefold. There exists a unique E_2-centre automorphism*

$$\sigma_Q \in \mathrm{Aut}_{\mathrm{E}_2}\big(Z_{\mathrm{E}_2}(\mathcal{F}_Q)\big)$$

whose action on partition-function classes effects the substitution $-q = e^{iu}$, intertwining M_{DT} and M_{PT}:

$$\sigma_Q \cdot Z_{\mathrm{DT}}(q) = Z_{\mathrm{PT}}(q) \cdot M(-q)^{\chi(Q)}, \qquad \chi(Q) = -200,$$

where $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the MacMahon function. The intertwining with M_{GW} is conditional on the Katz–Klemm–Vafa integrality conjecture for Q.

Proof sketch. Existence of σ_Q on the DT/PT pair is the Bridgeland–Toda wall-crossing identity (Bridgeland 2011, Toda 2010), valid for every smooth projective CY_3 and a fortiori for Q. The Kontsevich–Soibelman wall-crossing formula (KS 2008) provides the chain-level lift: the DT/PT ratio is the motivic MacMahon factor, and the substitution $-q = e^{iu}$ is the unique reparametrisation of the Novikov torus compatible with the E_2-factorisation structure on $Z_{\mathrm{E}_2}(\mathcal{F}_Q)$. Uniqueness

follows from the rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$ of a simply-connected E_2-algebra: $\pi_0\mathrm{Aut}_{\mathrm{E}_2}(Z_{\mathrm{E}_2}(\mathcal{F}_Q)) \hookrightarrow \mathrm{Aut}_{\mathrm{gr}}(H^*Z)$ and the graded automorphism group of the DT/PT vertex algebra contains a unique element effecting the MacMahon rescaling.

The GW leg requires KKV: Gromov–Witten invariants are identified with BPS counts via $\mathrm{GW}_g(Q) = \sum_\beta n^g_\beta \cdot$ (multiple cover), and the MNOP identity $Z_{\mathrm{GW}}(u) = Z_{\mathrm{PT}}(q)|_{-q=e^{iu}}$ holds at Q assuming the integrality of n^g_β. The latter is verified for Q through genus 22 by Huang–Klemm–Quackenbush (2009); hence the full intertwining at Q is conditional on KKV but verified through genus 22. □ □

Remark 1.2. The substitution $-q = e^{iu}$ is not a tautology. The three modules $M_{\mathrm{DT}}, M_{\mathrm{PT}}, M_{\mathrm{GW}}$ are distinct dualisable objects with a priori independent partition functions; the rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$ forces the intertwiner uniquely.

Remark 1.3 (Generic compact CY_3, via Joyce–KPS vertex algebra). Theorem **??** extends to any smooth projective CY_3 X: the Joyce vertex algebra V_X on the derived moduli stack $\widetilde{\mathbb{M}(r,c)}_X$ of perfect complexes (Joyce 2021, arXiv:2111.04694 Theorem 5.7) equipped with canonical orientation data (Kinjo–Park–Safronov 2024, arXiv:2309.08673 Theorem 1.2) carries three dualisable modules $M^X_{\mathrm{DT}}, M^X_{\mathrm{PT}}, M^X_{\mathrm{GW}}$ whose Behrend-weighted E_2-traces are intertwined by a unique E_2-centre automorphism σ_X effecting $-q = e^{iu}$. Uniqueness is Dunn additivity $\mathrm{Aut}_{\mathrm{E}_2}(Z_{\mathrm{E}_2}(V_X)) = \Omega^2\mathrm{Aut}(V_X)$: the graded-automorphism group is the Picard-lattice isometry group together with the KPS orientation $\mathbb{Z}/2$, forcing $\sigma_X : y \mapsto -y$, $(q,Q) \mapsto (q,Q)$. The DT/PT leg is unconditional; the PT/GW leg is KKV-conditional (known for toric per MNOP 2006, for primitive-fibre $K3 \times E$ per Oberdieck–Pandharipande 2015; projective-β case Maulik–Toda 2018). See Vol III Theorem `thm:mnop-generic-cy3-joyce-kps` (chapters/examples/k3_chiral_bialgebra_platonic.tex, subsection `subsec:mnop-generic-cy3-jo` for the complete statement.

2. Chain-level S^3-framing on the quintic

Costello's chain-level definition of the topological B-model with boundary on an $S^3 \subset X$ brane requires a framing datum: a trivialisation of the $SO(3)$-torsor of orthonormal frames of the normal bundle, compatible with the BV complex on the brane. At a Lagrangian $L \cong S^3 \hookrightarrow Q$ in the smooth quintic — e.g. the real locus of a suitable real deformation (Sheridan 2016; Sheridan–Smith 2021) — the framing obstruction lies in

$$\pi_0\big(\mathrm{Map}(S^3, SO(3))\big) = \pi_3(SO(3)) = \mathbb{Z},$$

corresponding to Hirzebruch's σ-invariant.

Proposition 2.1 (Canonical chain-level trivialisation)**.** *For a Lagrangian $L \cong S^3 \subset Q$ in a smooth quintic threefold, the framing torsor* $\mathrm{Fr}(L)$ *admits a canonical chain-level trivialisation induced by the CY_3 structure on Q. Explicitly, the line bundle*

$$\det T_L \otimes \det N_{L/Q} = \det\big(TQ|_L\big)$$

is canonically trivialised by the nowhere-vanishing holomorphic volume form $\Omega_Q \in H^0(Q, K_Q)$ (which trivialises K_Q since $c_1(Q) = 0$).

Proof. Q has $K_Q = \mathcal{O}_Q$ and Ω_Q is nowhere zero. Restriction to L gives a nowhere-vanishing section of $\det(TQ|_L)$; the Lagrangian splitting $TQ|_L = TL \oplus N_{L/Q}$ (as real bundles, under the complex structure) identifies $\det(TQ|_L)_{\mathbb{R}}$ with $\det(T_L) \otimes \det(N_{L/Q})$. Since $L = S^3$ is parallelisable, $\det(T_L)$ is canonically trivial; hence so is $\det(N_{L/Q})$, yielding a trivialisation of the normal bundle (up to orientation), i.e. a chain-level framing in the sense of Costello 2017 §4. □ □

Remark 2.2. Proposition **??** upgrades the Priority-7 assertion "CY-A topological ⇒ chain" on the quintic: the CY-A datum (holomorphic volume form) directly produces the chain-level framing datum without further contractible-choice input.

3. Chiral-volume conjecture at small enumerative degree

The chiral-volume conjecture (Vol III working notes, `conj:chiral-volume`) predicts, for a compact CY_3 X with a fixed rational-curve class $\beta \in H_2(X, \mathbb{Z})$,

$$\lim_{N\to\infty} \frac{1}{N} \log\big|Z_N(X, \beta)\big| = \frac{1}{2\pi} \big|\mathbf{AJ}(C_\beta)\big|,$$

where Z_N is the degree-N rational-curve count and $\mathbf{AJ}(C_\beta)$ is the Abel–Jacobi period of a representative curve. For the quintic Q and the hyperplane class $H \in H_2(Q)$, the degree-N genus-0 BPS invariant n_N is canonical (Candelas–de la Ossa–Green–Parkes 1991 through

$N = 3$; Kontsevich–Givental WDVV through $N = 5$; Huang–Klemm–Quackenbush 2007 BCOV anomaly recursion through $N = 10$):

$$\begin{aligned}
n_1 &= 2875\\
n_2 &= 609\,250\\
n_3 &= 317\,206\,375\\
n_4 &= 242\,467\,530\,000\\
n_5 &= 229\,305\,888\,887\,625\\
n_6 &= 248\,249\,742\,118\,022\,000\\
n_7 &= 295\,091\,050\,570\,845\,659\,250\\
n_8 &= 375\,632\,160\,937\,476\,603\,550\,000\\
n_9 &= 503\,840\,510\,416\,985\,243\,645\,106\,250\\
n_{10} &= 704\,288\,164\,978\,454\,686\,113\,488\,249\,750.
\end{aligned}$$

The sequence $a_N := N^{-1}\log n_N$ attains its minimum at $N = 3$:

$$a_1 \approx 7.964,\ a_2 \approx 6.660,\ a_3 \approx 6.525,\ a_4 \approx 6.554,\ a_5 \approx 6.613,$$

$$a_6 \approx 6.676,\ a_7 \approx 6.733,\ a_8 \approx 6.785,\ a_9 \approx 6.832,\ a_{10} \approx 6.873.$$

The sequence decreases sharply through $N = 3$, then increases monotonically: $a_{N+1} - a_N > 0$ for $N \in \{3, \dots, 9\}$, with increments $\{+0.029, +0.060, +0.062, +0.058, +$

Proposition 3.1 (Extended quintic chiral-volume data through $N = 10$)**.** *The sequence $a_N = N^{-1}\log n_N$ on the quintic with hyperplane class H has a minimum at $N = 3$ and, on the tail $N \in \{5, \dots, 10\}$, fits the three-parameter ansatz*

$$a_N \;=\; L \;+\; \alpha\,\frac{\log N}{N} \;+\; \frac{\beta}{N}$$

to residuals below 5×10^{-6} with $L = 7.593$, $\alpha = -3.32$, $\beta = 0.45$. The extrapolated limit L agrees to three significant figures with the Hodge-normalised mirror period $|t(z_c)|$ at the conifold point $z_c = 5^{-5}$ of the Candelas–de la Ossa–Green–Parkes mirror family:

$$|t(z_c)| \;=\; \left|\frac{\omega_1}{\omega_0}(z_c)\right| \;=\; 7.599,$$

with $\omega_0 = \sum_n ((5n)!/(n!)^5)\, z^n$ the fundamental period and $\omega_1 = \omega_0(z)\log z + \sum_n ((5n)!/(n!)^5) \cdot 5(H_{5n} - H_n)\, z^n$ the logarithmic period (H_k the k-th harmonic number). The attractor upper bound is the OSV rate $5\log 5 \approx 8.047$; the data $\{a_N\}$ rises monotonically from $N = 4$ toward this bound.

Proof. Genus-0 BPS invariants through $N = 10$ are the Huang–Klemm–Quackenbush table (*arXiv:0704.2440*, Table 1); the entries $N \le 5$ coincide with the direct Schubert/Katz/Ellingsrud–Strømme/Kontsevich/Givental

enumerative counts. Linear least-squares on the six equations $a_N = L + \alpha(\log N)/N + \beta/N$ for $N \in \{5, \dots, 10\}$ gives the stated coefficients with residuals bounded by 5×10^{-6}. The CDGP fundamental and logarithmic periods satisfy the Picard–Fuchs operator $\theta^4 - 5z(5\theta + 1)(5\theta + 2)(5\theta + 3)(5\theta + 4)$ with $\theta = z\partial_z$; evaluated at $z_c = 5^{-5}$ via the convergent power series (radius of convergence 5^{-5}, boundary value by continuity from the interior),

$$\omega_0(z_c) \approx 1.0696, \qquad \omega_1(z_c) \approx -8.128,$$

giving $|t(z_c)| \approx 7.599$. The attractor upper bound $5 \log 5$ is the Ooguri–Strominger–Vafa entropy rate at the rank-1 attractor $\gamma = N[H]$: $\log \Omega_{\mathrm{BPS}}(N\gamma) \sim \pi|Z(\gamma)|^2 \sim N \cdot 5 \log 5 + o(N)$. □ □

Remark 3.2. The extended data closes the $6 \le N \le 10$ computational gap of the prior small-N table. Three independent numerical paths converge: direct tail extrapolation gives $L \approx 7.59$, the CDGP mirror period $|t(z_c)| \approx 7.60$, the OSV attractor bound $5 \log 5 \approx 8.05$. The correction $\alpha = -3.32 < 0$ forces the approach to L from below: the chiral-volume limit is *not* bounded above by $\{a_1, a_2, a_3\}$; those points lie on the Schubert–Katz–Ellingsrud–Strømme Fano-geometric branch, below the large-N BPS regime that governs the asymptotic.

Remark 3.3 (Hodge normalisation of the Abel–Jacobi period). The conjecture $\lim a_N = (2\pi)^{-1}|\mathbf{AJ}(C_H)|$ uses the Hodge metric on $J^2(Q)$, equivalently (after Arakelov–Faltings) the Faltings norm $\|\omega\|_{\mathrm{Fal}}$ of the restriction of the holomorphic 3-form Ω_Q along C_H. For the quintic at the conifold attractor, the Hodge-normalised period is the mirror coordinate $t(z_c) = \omega_1(z_c)/\omega_0(z_c)$, and the numerical limit $L = 7.59$ equals $|t(z_c)| = 7.60$ directly: the $1/(2\pi)$ factor in the conjecture is absorbed into the $2\pi i$-normalised monodromy of the log-period ω_1. The three-path convergence on $L \in [7.59, 8.05]$ confirms this normalisation.

4. NCCR substitute at generic compact CY_3

Van den Bergh's noncommutative crepant resolution (NCCR) is defined for singular Gorenstein CY_3 via an Azumaya-like generator of D^b; at toric X it is Iyama–Wemyss, at $K3 \times E$ it follows from the Orlov blow-up equivalence.

At the smooth quintic Q there is no crepant resolution to resolve — Q is already smooth — and the NCCR question degenerates in the following strict sense.

Proposition 4.1 (Serre-equivariant quasi-NCCR on smooth compact CY_3)**.** *Let X be a smooth compact Calabi–Yau threefold. The Serre*

functor $S_X : D^b(X) \to D^b(X)$ satisfies $S_X = [3]$, and $D^b(X)$ equipped with this CY_3 structure is the Serre-equivariant quasi-NCCR for X: the full category $\mathrm{Perf}(X) = D^b(X)$ *plays the role of the NCCR generator* $\mathrm{End}_R(M)$*, with the Serre pairing providing the Van den Bergh duality datum.*

Proof. Smoothness plus $K_X = \mathcal{O}_X$ gives $S_X = \otimes K_X[3] = [3]$. The CY_3 structure provides the Calabi–Yau pairing

$$\mathrm{RHom}(F, G) \otimes \mathrm{RHom}(G, F) \to \mathbb{C}[-3]$$

which is precisely the Van den Bergh NCCR duality datum on the generator id of $\mathrm{Perf}(X)$. Kuznetsov's residual category $\mathrm{Res}(X)$ is the full $D^b(X)$ when X has trivial semiorthogonal decomposition, as holds for smooth CY_3 without exceptional objects (all $\mathrm{Hom}(F, F[k])$ nonzero in degrees $0, 3$ by Serre duality). □ □

Remark 4.2. Proposition **??** settles the Priority "global NCCR substitute" at the quintic cleanly: the substitute is the CY_3 structure itself. The nontrivial content of NCCR (desingularising a singular Gorenstein CY_3) is vacuous at smooth X; the Serre-equivariant structure is what survives and is what the Φ functor sees.

5. Honest-status summary

(1) MNOP $-q = e^{iu}$ as the unique E_2-centre automorphism on $Z_{\mathrm{E}_2}(\mathcal{F}_X)$: theorem on the (DT, PT) pair for every smooth compact CY_3 (Bridgeland–Toda + rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$); conditional on KKV for the (DT, PT, GW) triple. At the quintic, verified through genus 22.

(2) Chain-level S^3-framing on the quintic: theorem; canonical trivialisation by Ω_Q (Prop. **??**).

(3) Chiral-volume conjecture on the quintic through $N = 10$: the sequence $a_N = N^{-1} \log n_N$ has a minimum at $N = 3$ and increases monotonically thereafter; the three-parameter tail fit $a_N = L + \alpha(\log N)/N + \beta/N$ gives $L = 7.593$, matching the CDGP conifold-point mirror period $|t(z_c)| = 7.599$ to three significant figures and bounded above by the OSV attractor rate $5 \log 5 \approx 8.047$ (Prop. **??**).

(4) NCCR substitute at smooth compact CY_3: theorem; $D^b(X)$ with Serre $= [3]$ is the Serre-equivariant quasi-NCCR (Prop. **??**).

Items (1), (2), (4) upgrade prior conjectural statements to theorems within their scope; item (3) extends the quintic chiral-volume data through $N = 10$ and exhibits a three-path numerical convergence on

the limit $L \approx 7.59 \approx |t(z_c)|$ against the Candelas–de la Ossa–Green–Parkes conifold-point mirror period and the Ooguri–Strominger–Vafa attractor rate $5 \log 5$.